

PageRank Scientific Technical Report

Ashe Raymond-Barker
S2146569

1 Introduction

In 1990 Tim Berners-Lee developed the World Wide Web [1], making it possible to access information from anywhere in the world via the internet. The 1993 launch of the web browser Mosaic, led to an explosion of activity, as its user-friendly design opened the internet to a much wider audience. The number of webpages grew from around 130 in 1993 to over 100,000 by 1996 [2]

However, this expansion created an urgent challenge. With so many webpages emerging, search engines needed a way to determine which ones to recommend. Early engines such as Yahoo! relied primarily on human categorisation and keyword-based indexing, ranking pages by keyword frequency. However, they were easily manipulated and a practice known as Spamdexing - where popular keywords were excessively used in pages to artificially boost their ranking - became widespread [3]. Consequently, search results often became unreliable.

Content-based ranking systems were unable to effectively distinguish between useful information and keyword-heavy spam. There was a pressing need for something new. A system that considered the structural properties of the web, rather than just the content of webpages.

In 1998, Sergey Brin and Larry Page put forward PageRank [4], reframing web-search as a stochastic process. The web was modelled as a network of interconnected pages, forming a Markov chain in which a “random surfer” navigated links between pages. The stationary distribution of this process represented the long-term probability of the surfer being on a given page. Therefore, by finding this stationary distribution, each page would be given a quantitative measure of importance. The approach of capturing global page importance from local link structure was a conceptual breakthrough that combined graph theory, probability, and linear algebra.

While modern search engines incorporate more sophisticated techniques, PageRank remains foundational. Understanding its development and mathematical principles is essential to appreciate how link analysis revolutionised information retrieval.

2 The Development of PageRank

2.1 The Original Formula

Brin and Page proposed an initial formula to quantify the importance of webpages [5]:

$$\pi(x) = \sum_{y \in B_x} \frac{\pi(y)}{|y|} \quad (1)$$

where:

$\pi(x)$ is the PageRank (importance) of webpage x ,

B_x is the set of webpages that link to x ,

$\pi(y)$ is the PageRank of page y linking to x ,

$|y|$ is the number of outbound hyperlinks from page y .

However, this formula cannot be solved directly because the PageRank values of the inlinking pages, $\pi(y)$, are themselves unknown. To address this, an iterative approach is used. All pages are initially assigned the same PageRank:

$$\pi_0(x) = \frac{1}{n}, \quad x = 1, \dots, n, \quad (2)$$

where n is the total number of pages in the system. The PageRank is then updated iteratively:

$$\pi_{t+1}(x) = \sum_{y \in B_x} \frac{\pi_t(y)}{|y|} \quad (3)$$

After sufficient iterations, and under certain conditions discussed later, the values of $\pi(x)$ converge to a unique stationary distribution, providing a consistent measure of page importance across the network.

2.2 Mapping Webpages as Markov Chains

An intuitive way to understand the PageRank model and its formulation is to represent the web as a directed graph.

Definition 1 (Directed Graph). *A directed graph is an ordered pair*

$$G = (V, E)$$

where:

- V is a non-empty set whose elements are called vertices or nodes.
- $E \subseteq V \times V$ is a set of directed edges, which are ordered pairs (x, y) . Each directed edge goes from node x (the tail) to node y (the head).

[6]

In the context of the web, nodes correspond to webpages, and directed edges represent hyperlinks from one page to another.

Consider a system of five linked webpages:

Webpage	Webpage it links to
A	B, C, D
B	A, E
C	B
D	C, E
E	A, B

Table 1: Example of nodes and their outgoing hyperlinks

This system can be visualised as a directed graph:

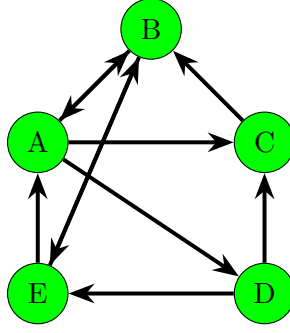


Figure 1: Directed graph of webpages A–E

To develop our intuition, consider the following analogy. When choosing a film to watch, friend A makes 10 recommendations, while friend B makes only 2. All else being equal, friend B’s recommendations clearly carry more weight, as their attention has been split across 2 films, whilst friend A’s has been split across 10. The same intuition applies to hyperlinks: a hyperlink from a page with many outgoing links should contribute less weight to each target than a hyperlink from a page with few outgoing links.

Formally, if page x has $|x|$ outgoing links, the probability that a random surfer on page x follows a link to page y is:

$$\mathbb{T}(x, y) = \begin{cases} \frac{1}{|x|} & \text{if } \exists \text{ a link from page } x \text{ to page } y \\ 0 & \text{otherwise} \end{cases}$$

These probabilities then form the Hyperlink matrix, H . Each row and column of H corresponds to a node (webpage) in the graph. A directed edge (hyperlink) from node x to node y corresponds to a positive entry $\mathbb{T}(x, y) > 0$. If there is no link, the entry is 0.

For the example given by Figure 1:

$$H = \begin{pmatrix} \mathbb{T}(A, A) & \mathbb{T}(A, B) & \mathbb{T}(A, C) & \mathbb{T}(A, D) & \mathbb{T}(A, E) \\ \mathbb{T}(B, A) & \mathbb{T}(B, B) & \mathbb{T}(B, C) & \mathbb{T}(B, D) & \mathbb{T}(B, E) \\ \mathbb{T}(C, A) & \mathbb{T}(C, B) & \mathbb{T}(C, C) & \mathbb{T}(C, D) & \mathbb{T}(C, E) \\ \mathbb{T}(D, A) & \mathbb{T}(D, B) & \mathbb{T}(D, C) & \mathbb{T}(D, D) & \mathbb{T}(D, E) \\ \mathbb{T}(E, A) & \mathbb{T}(E, B) & \mathbb{T}(E, C) & \mathbb{T}(E, D) & \mathbb{T}(E, E) \end{pmatrix} = \begin{pmatrix} 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 \\ \frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \end{pmatrix}$$

Note that H is the matrix representation of a *Probabilistic Directed Graph*, which forms the basis of PageRank.

Definition 2 (Probabilistic Directed Graph (PDG)). *A PDG is a directed graph in which edges have probabilities corresponding to the likelihood of moving from one node to another.*

Definition 3 (Stochastic Matrix). *A Stochastic matrix is a square matrix M whose entries are non-negative such that the sum of the entries in each row equals 1.*

$$M_{ij} \geq 0, \quad \sum_j M_{ij} = 1 \quad \forall i \quad (4)$$

[7]

Definition 4 (Markov Chain). *A stochastic process $\{X_t\}$ on a state space S is called a Markov chain if for all t and all states $i, j \in S$,*

$$\mathbb{P}(X_{t+1} = j \mid X_t = i, X_{t-1}, \dots, X_0) = \mathbb{P}(X_{t+1} = j \mid X_t = i). \quad (5)$$

[8]

The Hyperlink matrix for our example is clearly stochastic (4), ensuring that each row represents a valid probability distribution over the next nodes. When this matrix is used to define the transitions in a stochastic process X_n , the probability of moving from node x to state y depends only on the current node. Hence, using a row-stochastic matrix as the transition matrix guarantees that the process satisfies the Markov property (5), making H the transition matrix of a discrete-time Markov chain.

2.3 Vector Problems. formation of distribution to stationary distribution

Given that the web can be represented as a Markov chain with transition matrix H , we can now examine how a vector, representing the PageRanks of all webpages, evolves over time to reach a stationary distribution.

Consider the probability of a surfer being at a given webpage at a given time. We first initialise a PageRank vector to store the PageRank of n pages at time t :

$$\Pi_t = [\pi_t(0), \pi_t(1), \dots, \pi_t(n)]$$

Applying the PageRank formula (2) (3) to the example in Figure 1, we have:

$$\Pi_0 = [\frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5}], \quad \Pi_t = [\pi_t(A), \pi_t(B), \pi_t(C), \pi_t(D), \pi_t(E)]$$

Then the PageRank of page A is calculated by:

$$\begin{aligned} \pi_{t+1}(A) &= \pi_t(A) \mathbb{T}(A, A) \\ &\quad + \pi_t(B) \mathbb{T}(B, A) \\ &\quad + \pi_t(C) \mathbb{T}(C, A) \\ &\quad + \pi_t(D) \mathbb{T}(D, A) \\ &\quad + \pi_t(E) \mathbb{T}(E, A) \end{aligned}$$

where $\mathbb{T}(x, y)$ is the probability of a surfer stepping from node x to node y . That is, clicking a hyperlink on webpage x , that takes them to webpage y .

This can be written as vector multiplication:

$$\pi_{t+1}(A) = \Pi_t * T_{i,A} \quad i \in V$$

where

$$T_{i,A} = \begin{bmatrix} \mathbb{T}(A, A) \\ \mathbb{T}(B, A) \\ \mathbb{T}(C, A) \\ \mathbb{T}(D, A) \\ \mathbb{T}(E, A) \end{bmatrix}$$

This is equivalent to the first column of the Hyperlink matrix, H . By calculating the PageRank of the other pages with this formula, we notice that we can rewrite the equations as a matrix multiplication:

$$\Pi_{t+1} = \Pi_t * H \tag{6}$$

where H is the Hyperlink matrix we previously defined.

By repeatedly multiplying by H , the values of PageRank, Π , will converge to a *stationary distribution*.

Definition 5 (Stationary Distribution). *A distribution Π for a Markov chain with transition matrix H is called a stationary distribution if*

$$\Pi = \Pi H, \tag{7}$$

with the constraint that

$$\sum_i \pi_i = 1 \quad \text{and} \quad \pi_i \geq 0 \text{ for all } i.$$

This stationary distribution represents the steady-state probabilities, meaning no matter how many times we apply the transition matrix, the distribution will remain the same. Thus, it provides a quantitative value of the importance of the webpages, where a higher value indicates a higher probability of a surfer being on that page at any given time, and thus a greater importance.

In the example in Figure 1, this method ranks the pages in order of importance as follows:

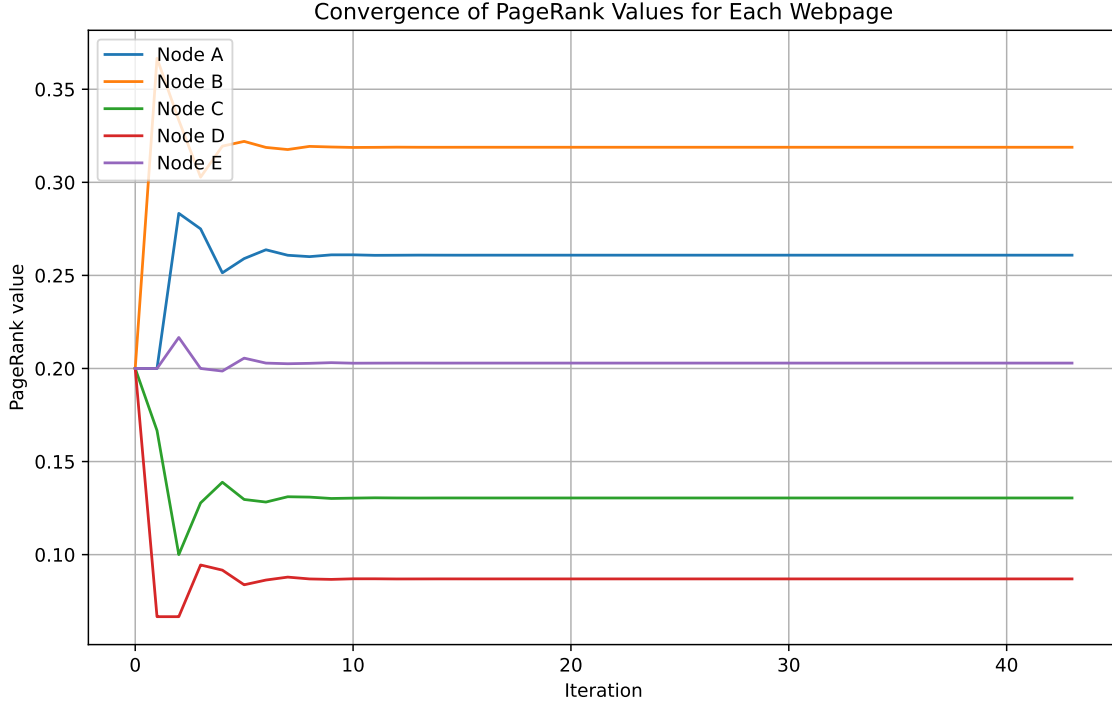


Figure 2: Convergence of PageRank Values in Example 1

Webpage	Final PageRank Value
<i>A</i>	0.260870
<i>B</i>	0.318841
<i>C</i>	0.130435
<i>D</i>	0.086957
<i>E</i>	0.202899

Table 2: Final PageRank Values for Example 1

2.4 The Fundamental Theorem of Markov Chains

For PageRank to work properly, the initial distribution, Π_0 , must converge to a unique stationary distribution. Uniqueness is essential because, otherwise, the “importance” or PageRank of a webpage could depend on the initial distribution, which would make the ranking ambiguous. Linn and Page encountered a key challenge: not all graphs naturally guarantee this convergence. However, given that a system of webpages could be modelled as a Markov Chain (5) (as the movement of a random web surfer depends only on their current webpage), Linn and Page were able to counter this problem by utilising *Fundamental Theorem of Markov Chains* to guarantee the existence of a stationary distribution. This theorem relied on Markov Chains being *irreducible*, *aperiodic* and *positive-recurrent*.

Definition 6. *Imagine starting at a random node (webpage), x on our web graph. Each step we follow a hyperlink to a new node. Then we denote the probability of being at node y after t steps as:*

$$\mathbb{T}^t(\mathbf{x}, \mathbf{y})$$

Definition 7 (Irreducible Markov Chain). *A Markov chain with state space V and transition matrix $H = [\mathbb{T}(x, y)]$ is irreducible iff for all nodes (webpages) $x, y \in V$, there exists an integer $t \geq 0$ such that*

$$\mathbb{T}^t(x, y) > 0.$$

That is, it is possible to reach any node from any other node in a finite number of steps. [10]

Definition 8 (Aperiodic Markov Chain). *A Markov chain with state space V and transition matrix $H = [\mathbb{T}(x, y)]$ is aperiodic iff for all nodes (webpages) $x \in V$, the greatest common divisor of the set of times $t \geq 1$ such that $\mathbb{T}^t(x, x) > 0$ is 1:*

$$\gcd\{t \geq 1 : \mathbb{T}^t(x, x) > 0\} = 1.$$

That is, there is no fixed cycle length for returning to any node. [10]

Definition 9 (Recurrent and Positive Recurrent Nodes). *Let a Markov chain have state space V and transition matrix $H = [\mathbb{T}(x, y)]$. A node (webpage) $x \in V$ is recurrent iff, starting from x , the probability of eventually returning to x is 1:*

$$P(\text{return to } x \mid X_0 = x) = 1.$$

[11]

A recurrent node x is positive-recurrent iff the expected return time to x is finite:

$$E[\mu_x] = \sum_{t=1}^{\infty} t \cdot \mathbb{T}^t(x, x) < \infty,$$

where X_t denotes the position of a random surfer at time t , and $\mu_x = \inf\{t > 0 : X_t = x \mid X_0 = x\}$ is the first return time to node x . [12]

Definition 10 (Positive Recurrent Markov Chain). *A Markov chain with state space V and transition matrix $H = [\mathbb{T}(x, y)]$ is positive-recurrent iff every node (webpage) in V is positive-recurrent*

It turns out that the example in Figure 1 satisfies all of these definitions. These definitions are essential for understanding and proving the *Fundamental Theorem of Markov Chains*. However, before we can prove that theorem, we must first establish the *Chapman-Kolmogorov Equation*.

Theorem 1. (Chapman-Kolmogorov Equation) *For all $k \in \mathbb{N} \cup \{0\}$,*

$$\mathbb{T}^t(x, y) = \sum_{z \in V} \mathbb{T}^k(x, z) \mathbb{T}^{t-k}(z, y) \quad (8)$$

[13]

Proof. We introduce an intermediate node (webpage), z , that the chain may occupy at some time, k , where $0 \leq k \leq t$ and apply the law of total probability:

$$\mathbb{T}^t(x, y) = \mathbb{P}(X_t = y \mid X_0 = x) = \sum_{z \in V} \mathbb{P}(X_t = y, X_k = z \mid X_0 = x)$$

Expanding using conditional probability gives:

$$\sum_{z \in V} \mathbb{P}(X_t = y, X_k = z \mid X_0 = x) = \sum_{z \in V} \mathbb{P}(X_t = y \mid X_k = z, X_0 = x) \mathbb{P}(X_k = z \mid X_0 = x)$$

By the Markov Property, the future depends only on the present, so

$$\sum_{z \in V} \mathbb{P}(X_t = y \mid X_k = z, X_0 = x) = \sum_{z \in V} \mathbb{P}(X_t = y \mid X_k = z)$$

Then by recognising the that:

$$\begin{aligned} \mathbb{P}(X_t = y \mid X_k = z) &= \mathbb{T}^{t-k}(z, y) \\ \mathbb{P}(X_k = z \mid X_0 = x) &= \mathbb{T}^k(x, z) \end{aligned}$$

We obtain the Chapman-Kolmogorov equation:

$$\mathbb{T}^t(x, y) = \sum_{z \in V} \mathbb{T}^k(x, z) \mathbb{T}^{t-k}(z, y)$$

□

Finally, we are ready to prove the *Fundamental Theorem of Markov Chains*. This theorem guided the adaptation of the Hyperlink matrix to ensure that a unique stationary distribution is always guaranteed, thereby allowing PageRank to consistently and effectively rank the importance of webpages.

Theorem 2. (*Fundamental Theorem of Markov Chains*) For any irreducible, aperiodic, and positive recurrent Markov chain, there exists a unique stationary distribution, $\Pi = \{\pi(x) : x \in V\}$, such that for all $x \in V$,

$$\pi(x) = \sum_{y \in V} \pi(y) \mathbb{T}(y, x), \quad (9)$$

and

$$\sum_{y \in V} \pi(y) = 1, \quad \pi(y) \geq 0 \quad \forall y \in V.$$

[13]

Proof. Given that the chain is irreducible, aperiodic and positive-recurrent, the long-term probability of being at any node (webpage) is strictly positive.

$$\forall y \in V, \quad \pi(y) = \lim_{t \rightarrow \infty} \mathbb{T}^t(x, y) > 0$$

where $\mathbb{T}^t(x, y)$ represents the probability of transitioning from node x to node y in t steps.

As Π is a probability distribution, it follows that $\sum_{y \in V} \pi(y) = 1$

For any $t, M \in \mathbb{Z}^+$, we know that:

$$\sum_{x=0}^M \mathbb{T}^t(x, y) \leq \sum_{x=0}^{\infty} \mathbb{T}^t(x, y) \leq 1$$

Taking the limits as $t, M \rightarrow \infty$ we get:

$$\lim_{t, M \rightarrow \infty} \sum_{x=0}^M \mathbb{T}^t(x, y) = \sum_{x=0}^{\infty} \pi_y \leq 1 \implies \text{For any } M, \quad \sum_{x=0}^M \pi_y \leq 1$$

Now we apply the Chapman-Kolmogorov equation (8):

$$\mathbb{T}^{t+1}(x, y) = \sum_{z=0}^{\infty} \mathbb{T}^t(x, z) \mathbb{T}(z, y) \geq \sum_{z=0}^M \mathbb{T}^t(x, z) \mathbb{T}(z, y)$$

Taking the limits as $t, M \rightarrow \infty$, we obtain:

$$\pi_y \geq \sum_{x=0}^{\infty} \pi_z \mathbb{T}(z, y)$$

Now we assume that a strict inequality holds for some node y :

$$\pi_y > \sum_z \pi_z \mathbb{T}(z, y) \quad \text{for at least one } y \in V$$

Summing this inequality over all nodes, we get:

$$\sum_y \pi_y > \sum_y \sum_z \pi_z \mathbb{T}(z, y) = \sum_y \pi_z \sum_y \mathbb{T}(z, y)$$

Note that $\sum_y \mathbb{T}(z, y)$ is the probability of leaving node z and transitioning to any other node in 1 step, which happens with certainty. So

$$\sum_y \pi_z \sum_y \mathbb{T}(z, y) = \sum_y \pi_z$$

Note that $\sum_y \pi_z$ is a probability distribution and thus equals 1 Hence we have:

$$\begin{aligned} \sum_y \pi_z \sum_y \mathbb{T}(z, y) &= 1 \\ \implies \sum_y \pi_y &> 1 \end{aligned}$$

But this is a contradiction as we know that the total probability cannot exceed 1. Hence, the assumption that a strict inequality holds is impossible meaning equality must hold. So:

$$\pi_y = \sum_{z \in V} \pi_z \mathbb{T}(z, y)$$

A unique stationary distribution exists. □

Therefore, to guarantee that PageRank works, we must ensure that our Hyperlink matrix is formed in such a way that its corresponding Markov chain is always irreducible, aperiodic and positive recurrent.

2.5 Dangling Nodes

We can now examine cases where our current formation of the Hyperlink matrix fails to meet the necessary conditions for convergence to a unique stationary distribution. This will enable us to develop it to consistently satisfy the required properties moving forward.

Consider the following example of a web and its Hyperlink matrix:

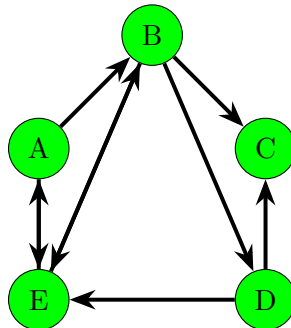


Figure 3: Directed graph of webpages with a *Dangling Node*

$$H = \begin{pmatrix} 0 & \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 0 & 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \end{pmatrix}$$

Definition 11. (*Dangling Node*) A dangling node is a node (webpage) with no outlinking edges (hyperlinks)

Webpage C is a dangling node, reflected in the transition matrix by a row of zeros. A state with no outgoing transitions is problematic: if a surfer lands on this page, they are stuck indefinitely. Hence, some parts of the chain become unreachable from others, making the Markov chain reducible. Consequently, there is no longer a unique stationary distribution.

Fortunately, this problem has an easy fix. If you were to reach a webpage with no hyperlinks, you would probably open a new window and visit a random page. We represent this mathematically by replacing the zeros in the dangling node's row with a uniform probability distribution across all nodes. This models the probability of you opening up a new page with uniform likelihood.

Formally, let H be the original hyperlink matrix and define a vector a such that:

$$a_x = \begin{cases} 1 & \text{if node } x \text{ is a dangling node} \\ 0 & \text{otherwise} \end{cases}$$

Let e^T be a row vector of all ones, and let n be the total number of nodes. The modified Hyperlink Stochastic matrix is then:

$$S = H + \frac{1}{n}ae^T \quad (10)$$

This adjustment ensures that every time a surfer reaches a dangling node, they are randomly redirected to another node in the chain. Now, there are no states where a surfer can become permanently stuck. All nodes are reachable and the Markov chain is once again irreducible.

We now have matrix for the example in Figure 3 that will give us a unique stationary distribution that allows us to quantify page importance:

$$S = \begin{pmatrix} 0 & \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 0 & 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} \\ 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \end{pmatrix}$$

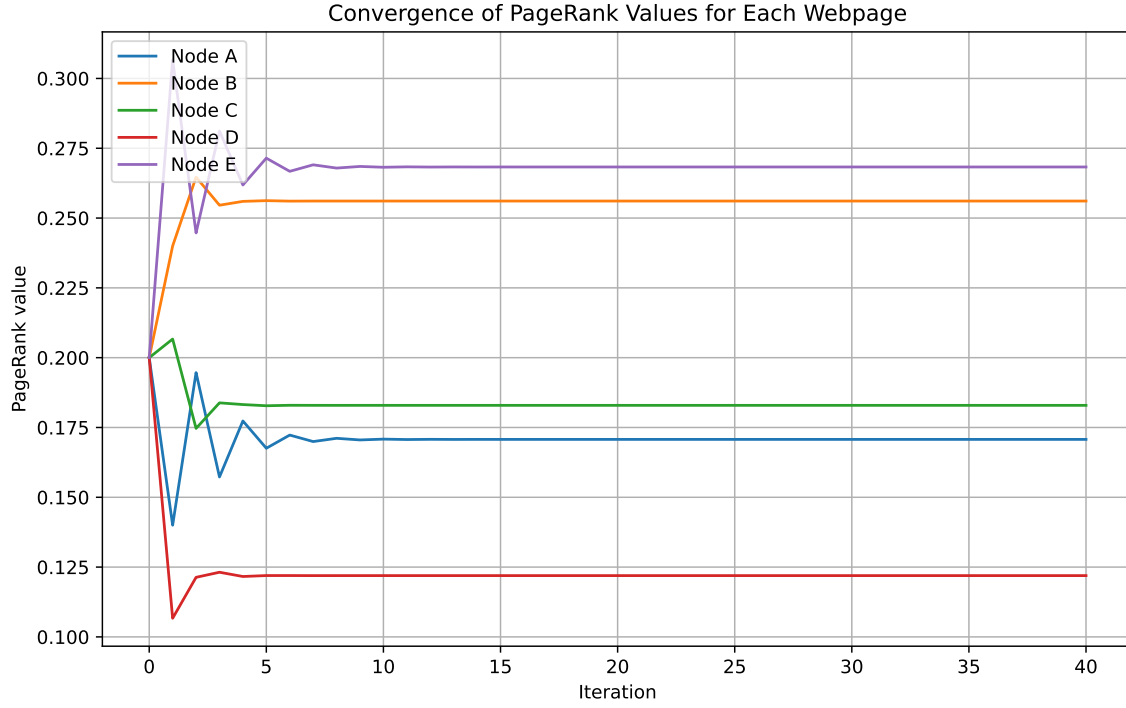


Figure 4: Convergence of PageRank Values in Example 2

Webpage	Final PageRank Value
<i>A</i>	0.170732
<i>B</i>	0.256098
<i>C</i>	0.182927
<i>D</i>	0.121951
<i>E</i>	0.268293

Table 3: Final PageRank Values for Example 2

2.6 Periodic Behaviour and Distinct Chains and the google matrix

Now imagine we have the following graph of webpages and its corresponding Hyperlink matrix:

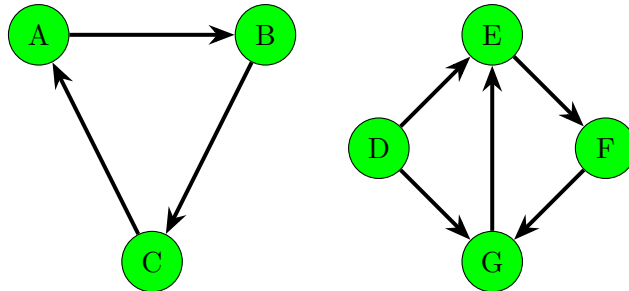


Figure 5: Directed graph of webpages with periodic behaviour and distinct chains

$$H = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix}$$

This structure presents significant issues. First, if a random surfer begins in one cluster, the other cluster cannot be reached, causing those pages to have zero rank. Second, the cyclic connections within each cluster render the Markov chain periodic, preventing the guarantee of a unique stationary distribution. In addition, whilst there are no dangling nodes, some nodes may have no incoming links, raising the question of how a surfer could ever reach these nodes.

To address these problems, we adopt the principle that a surfer may, at any point, abandon their current path and jump to a randomly selected webpage. Brin and Page formalized this idea by combining the Hyperlink Stochastic matrix S with a primitive Teleportation matrix representing uniform random teleportation,

$$J = \frac{1}{n}ee^T,$$

where n is the total number of pages and e is a vector of ones. This yields the Google Matrix:

$$G = \alpha S + (1 - \alpha)J = \alpha H + \frac{\alpha}{n}ae^T + \frac{1 - \alpha}{n}ee^T, \quad (11)$$

where $\alpha \in (0, 1)$ reflects the probability that a surfer follows a hyperlink rather than teleporting randomly, H is the original Hyperlink matrix, and a indicates dangling nodes. Google reportedly sets $\alpha \approx 0.85$ [14].

This modification ensures that every page in the network is reachable from every other page, making the chain irreducible. The chain is also aperiodic due to random teleportation breaking cycles. Furthermore, the chain is positive recurrent, and thus, by the Fundamental Theorem of Markov Chains, a unique stationary distribution exists.

The Google Matrix for the example in Figure 5 allows us to calculate a stationary distribution and thus rank the importance of the pages (using $\alpha = 0.85$):

$$G = \alpha \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix} + \frac{1 - \alpha}{7} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix}$$

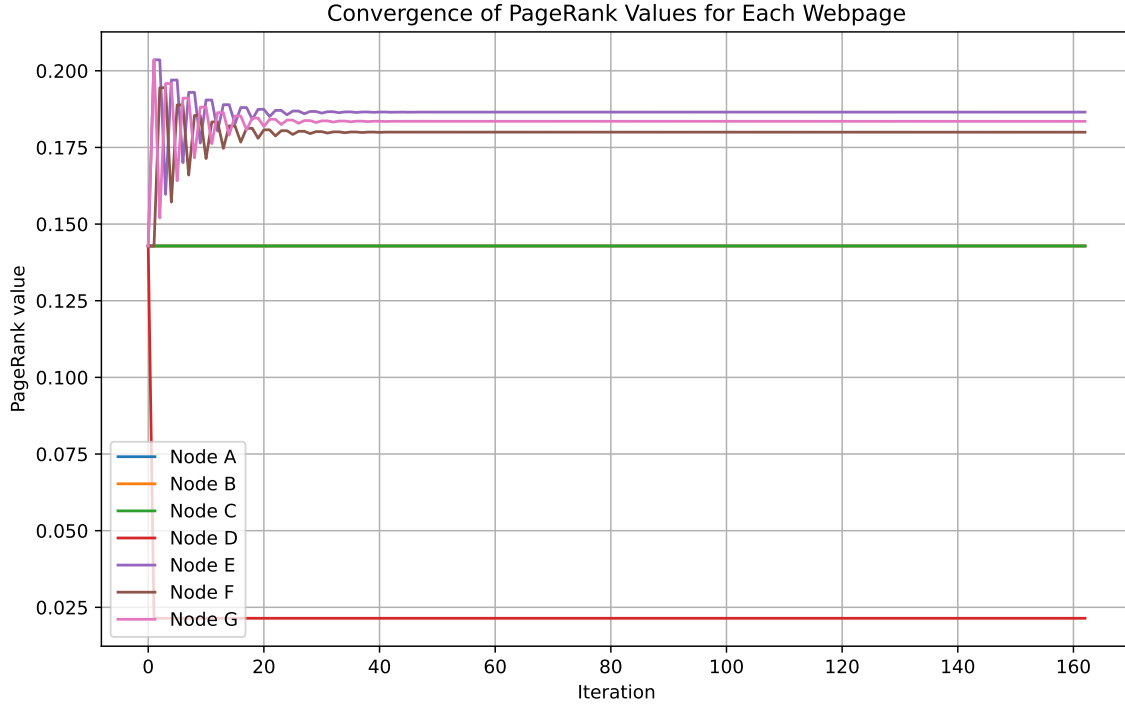


Figure 6: Convergence of PageRank Values in Example 3

Webpage	Final PageRank Value
<i>A</i>	0.142857
<i>B</i>	0.142857
<i>C</i>	0.142857
<i>D</i>	0.021429
<i>E</i>	0.186520
<i>F</i>	0.179970
<i>G</i>	0.183510

Table 4: Final PageRank Values for Example 3

2.7 A Note on Positive-Recurrence

For any finite Markov chain, irreducibility guarantees positive-recurrence by the *Finite Hitting Theroem* [10]. This is because, in an irreducible finite Markov chain, there is a path between every pair of nodes, meaning that all nodes are recurrent (i.e., the chain will eventually return to any starting node). Moreover, every recurrent node must also be positive-recurrent. This is because the probability of returning to a node is 1, and since there are only finitely many nodes, the expected return time must be finite. Consequently, as the number of webpages - though large - is finite, ensuring irreducibility and aperiodicity is sufficient to guarantee the existence of a unique stationary distribution.

2.8 Improving the Google Matrix

The Google Matrix has since been adapted in many ways to better represent the actual browsing behaviour of surfers. This adaptation is crucial in ensuring that the model most accurately reflects real-world internet

usage, thereby improving its recommendations of webpages.

One key modification addresses the fact that, in reality, surfers do not randomly jump from one webpage back onto the same webpage. They are also unlikely to randomly jump to any random webpage with uniform probability. Instead, they tend to jump to a related page, or something similar. To account for this, we redefine the Teleportation matrix as follows:

$$j(x, y) = \begin{cases} 0 & \text{if } x = y, \\ w_x & \text{if } x \neq y, \end{cases}$$

where w_x is a normalized weight such that

$$\sum_{j \neq i} w_x = 1.$$

This modified teleportation matrix enables more realistic browsing behavior, where w_x is distributed in a way that reflects the similarity of webpages and $j(x, y)$ ensures that a surfer will not jump back to a page they are currently on. As a result, the surfer is more likely to navigate to a related or similarly ranked page, leading to a more accurate and realistic calculation of PageRank.

3 conclusion

As of October 11, 2025, Alphabet’s (Google’s parent company) market valuation stands at \$2.921 trillion [15]. This extraordinary success is rooted in the development and application of the PageRank algorithm. Google’s application of mathematics to solve a real-world problem – the organization of information in an efficient and user-centric way – powered not only search results but also Google’s ability to generate trillions of dollars from targeted advertising. Moreover, Google’s success has revolutionized the way people communicate, interact, learn, and access information. The success of PageRank is a testament to the power of innovation and continues to shape the digital landscape in ways that were once unimaginable.

References

- [1] World Wide Web Consortium, *A Little History of the World Wide Web*, Accessed 09 October 2025, <https://www.w3.org/History.html>
- [2] National Science and Media Museum, *A Short History of the Internet*, Published 3 December 2020, <https://www.scienceandmediamuseum.org.uk/objects-and-stories/short-history-internet>
- [3] Webspam.org, *SEO Spam: What is Spamdexing?*, <https://www.webspam.org/seo-spam-what-is-spamdexing/>
- [4] L. Page, S. Brin, R. Motwani, and T. Winograd, *The PageRank Citation Ranking: Bringing Order to the Web*, Stanford University, Technical Report 1999-66, 1999, <http://ilpubs.stanford.edu:8090/422/1/1999-66.pdf>
- [5] S. Carrales, *Mathematics for Industry Case Study 1 - Google PageRank*, 2025.
- [6] A. Brandt, *Directed Graphs*, <https://www.csd.uwo.ca/~abrandt5/teaching/DiscreteStructures/Chapter7/directedgraphs.html>
- [7] Georgia Institute of Technology, *Stochastic Matrices*, <https://textbooks.math.gatech.edu/ila/stochastic-matrices.html>

- [8] Durham University, *Markov Chains Lecture Notes (MATH2647)*, https://maths.dur.ac.uk/stats/courses/ProbMC2H/_files/handouts/1516MarkovChains2H.pdf
- [9] Guangliang Chen (San José State University), *Stationary Distributions and Limiting Probabilities*, <https://www.sjsu.edu/faculty/guangliang.chen/Math263/lec4LimitingProb.pdf>
- [10] University of Cambridge, *Lecture 2: Probability Theory*, <https://www.cl.cam.ac.uk/teaching/1819/Probability/materials/Lecture2.pdf>
- [11] C. Casarotto, *Markov Chains and the Ergodic Theorem*, August 2007, <https://math.uchicago.edu/~may/VIGRE/VIGRE2007/REUPapers/FINALFULL/Casarotto.pdf>
- [12] Yibi (Department of Statistics, University of Chicago), *Lecture 5: [Title, if known]*, STAT 317 Lecture Notes, <https://www.stat.uchicago.edu/~yibi/teaching/stat317/2021/Lectures/Lecture5.pdf>
- [13] A. Plavnick, *The Fundamental Theorem of Markov Chains*, <https://www.math.uchicago.edu/~may/VIGRE/VIGRE2008/REUPapers/Plavnick.pdf>
- [14] Iain Gordon, *Honours Algebra 2024 Lecture Notes*, [file:///C:/Users/s2146569/Downloads/Honours_Algebra_2024_Lecture_Notes%20\(1\).pdf](file:///C:/Users/s2146569/Downloads/Honours_Algebra_2024_Lecture_Notes%20(1).pdf)
- [15] Macrotrends, *Alphabet Net Worth*, <https://www.macrotrends.net/stocks/charts/GOOGL/alphabet/net-worth>

A Appendix A

This appendix includes the code used to generate the graphs shown in Figures 2, 4 and 6, in addition to the calculation of PageRank's shown in Tables 2, 3 and 4.

```

1 import numpy as np
2 from numpy.linalg import norm
3 import pandas as pd
4 import matplotlib.pyplot as plt
5
6 # Example 1
7
8 n = 5; # Number of webpages
9
10 # Create Hyperlink Matrix
11 H = np.asarray([[0, 1/3, 1/3, 1/3, 0],
12                [1/2, 0, 0, 0, 1/2],
13                [0, 1, 0, 0, 0],
14                [0, 0, 1/2, 0, 1/2],
15                [1/2, 1/2, 0, 0, 0]])
16
17 maxiter = 1000000 # Max number of iterations
18 tol = 1e-12 # Tolerance value
19
20 # Initialize vector pi
21 piT = np.ones((1, n))/n
22
23 # Store piT values at each iteration
24 pi_history = [piT.flatten()]
25
26 for k in range(1, maxiter + 1):

```

```

27     piT_old = piT.copy()
28     # Apply an iteration of the Power Method
29     piT = piT @ H
30     pi_history.append(piT.flatten())
31
32
33     # Check for convergence of the method with 2-norm
34     if norm(piT-piT_old)/norm(piT_old) < tol:
35         break
36
37 if k == maxiter:
38     print('No convergence - max number of iterations reached!')
39
40 else:
41     # State number of iterations required
42     print('Convergence - number of iterations:', k)
43
44 # Convert history to numpy array for plotting
45 pi_history = np.array(pi_history)
46
47 # Plot convergence of each webpage
48 plt.figure(figsize=(10,6))
49 webpage_labels = ['A', 'B', 'C', 'D', 'E']
50 for i in range(n):
51     plt.plot(pi_history[:, i], label=f'Node_{webpage_labels[i]}')
52 plt.xlabel('Iteration')
53 plt.ylabel('PageRank value')
54 plt.title('Convergence of PageRank Values for Each Webpage')
55 plt.grid(True)
56 plt.legend()
57
58 # Save the plot as a PDF file
59 plt.savefig("example1.pdf", format="pdf", dpi=300) # Saves the plot as a PDF
60 plt.show()
61
62 # Present final PageRank vector in a table
63 df = pd.DataFrame({'Node': webpage_labels,
64                    'PageRank': piT.flatten()})
65 print("\nFinal PageRank Vector:")
66 print(df.to_string(index=False))
67
68
69 # Example 2
70
71 n = 5; # Number of webpages
72
73 # Create Stochastic Matrix
74 S = np.asarray([[0, 1/2, 0, 0, 1/2],
75                 [0, 0, 1/3, 1/3, 1/3],
76                 [1/5, 1/5, 1/5, 1/5, 1/5],
77                 [0, 0, 1/2, 0, 1/2],
78                 [1/2, 1/2, 0, 0, 0]])
79
80 maxiter = 1000000 # Max number of iterations
81 tol = 1e-12 # Tolerance value
82

```

```

83 # Initialize vector pi
84 piT = np.ones((1, n))/n
85
86 # Store piT values at each iteration
87 pi_history = [piT.flatten()]
88
89 for k in range(1, maxiter + 1):
90     piT_old = piT.copy()
91     # Apply an iteration of the Power Method
92     piT = piT @ S
93     pi_history.append(piT.flatten())
94
95
96     # Check for convergence of the method with 2-norm
97     if norm(piT-piT_old)/norm(piT_old) < tol:
98         break
99
100 if k == maxiter:
101     print('No convergence - max number of iterations reached!')
102
103 else:
104     # State number of iterations required
105     print('Convergence - number of iterations:', k)
106
107 # Convert history to numpy array for plotting
108 pi_history = np.array(pi_history)
109
110 # Plot convergence of each webpage
111 plt.figure(figsize=(10,6))
112 webpage_labels = ['A', 'B', 'C', 'D', 'E']
113 for i in range(n):
114     plt.plot(pi_history[:, i], label=f'Node_{webpage_labels[i]}')
115 plt.xlabel('Iteration')
116 plt.ylabel('PageRank value')
117 plt.title('Convergence of PageRank Values for Each Webpage')
118 plt.grid(True)
119 plt.legend()
120
121 # Save the plot as a PDF file
122 plt.savefig("example2.pdf", format="pdf", dpi=300) # Saves the plot as a PDF
123 plt.show()
124
125 # Present final PageRank vector in a table
126 df = pd.DataFrame({'Node': webpage_labels,
127                    'PageRank': piT.flatten()})
128 print("\nFinal PageRank Vector:")
129 print(df.to_string(index=False))
130
131
132 # Example 3
133
134 # Choose parameter alpha
135 alpha = 0.85;
136 print('alpha=', alpha)
137
138 n = 7; # Number of webpages

```

```

139
140 # Create Stochastic Matrix
141 S = np.asarray([[0, 1, 0, 0, 0, 0, 0],
142                [0, 0, 1, 0, 0, 0, 0],
143                [1, 0, 0, 0, 0, 0, 0],
144                [0, 0, 0, 0, 1/2, 0, 1/2],
145                [0, 0, 0, 0, 0, 1, 0],
146                [0, 0, 0, 0, 0, 0, 1],
147                [0, 0, 0, 0, 1, 0, 0]])
148
149 # Create Google Matrix
150 G = alpha * S + (1-alpha) / n * np.ones((n,n))
151
152 maxiter = 1000000 # Max number of iterations
153 tol = 1e-12 # Tolerance value
154
155 # Initialize vector pi
156 piT = np.ones((1, n))/n
157
158 # Store piT values at each iteration
159 pi_history = [piT.flatten()]
160
161 for k in range(1, maxiter + 1):
162     piT_old = piT.copy()
163     # Apply an iteration of the Power Method
164     piT = piT @ G
165     pi_history.append(piT.flatten())
166
167     # Check for convergence of the method with 2-norm
168     if norm(piT-piT_old)/norm(piT_old) < tol:
169         break
170
171 if k == maxiter:
172     print('No convergence - max number of iterations reached!')
173
174 else:
175     # State number of iterations required
176     print('Convergence - number of iterations: ', k)
177
178 # Convert history to numpy array for plotting
179 pi_history = np.array(pi_history)
180
181 # Plot convergence of each webpage
182 plt.figure(figsize=(10,6))
183 webpage_labels = ['A', 'B', 'C', 'D', 'E', 'F', 'G']
184 for i in range(n):
185     plt.plot(pi_history[:, i], label=f'Node_{webpage_labels[i]}')
186 plt.xlabel('Iteration')
187 plt.ylabel('PageRank value')
188 plt.title('Convergence of PageRank Values for Each Webpage')
189 plt.grid(True)
190 plt.legend()
191
192 # Save the plot as a PDF file
193 plt.savefig("example3.pdf", format="pdf", dpi=300) # Saves the plot as a PDF
194

```

```

195 plt.show()
196
197 # Present final PageRank vector in a table
198 df = pd.DataFrame({'Node': webpage_labels,
199                   'PageRank': piT.flatten()})
200 print("\nFinal PageRank Vector:")
201 print(df.to_string(index=False))

```

B Appendix B

Word count is below limit:

The screenshot shows an Overleaf document titled "PageRank Technical Report". A "Word count" pop-up window is displayed in the center, showing the following statistics:

- Total Words: 2726
- Headers: 11
- Math Inline: 106
- Math Display: 51

The document content includes LaTeX code for the PageRank matrix and a list of references. The word count window also displays several error messages:

- !!! Encountered \end without corresponding \begin !!!
- !!! Encountered \end without corresponding \begin !!!
- !!! Reached end of file while waiting for). !!!
- !!! Reached end of file while waiting for \end. !!!
- !!! Environment ended while waiting for \end(thebibliography). !!!
- !!! Reached end of file while waiting for \end. !!!
- !!! Environment ended while waiting for \end(document). !!!

The document content includes the following text:

This structure presents significant issues. First, the other cluster cannot be reached. Second, the cyclic connections within each cluster prevent the guarantee of a unique stationary distribution. If there are no dangling nodes, some nodes may have no incoming links, which means how a surfer could ever reach these nodes.

To address these problems, we adopt the principle of random teleportation. We abandon their current path and jump to a randomly selected webpage. We formalized this idea by combining the Hyperlink Stochastic matrix S with a primitive Teleportation matrix representing uniform random teleportation,

$$J = \frac{1}{n} \mathbf{1} \mathbf{1}^T + \alpha S$$

where n is the total number of pages and $\mathbf{1}$ is a vector of ones. This yields the Google Matrix:

$$G = \alpha S + (1-\alpha) J = \alpha S + \frac{(1-\alpha)}{n} \mathbf{1} \mathbf{1}^T + \alpha (1-\alpha) \mathbf{1} \mathbf{1}^T$$

where α in $(0,1)$ reflects the probability that a surfer follows a hyperlink rather than teleporting randomly, S is the original hyperlink matrix, and $\mathbf{1}$ indicates dangling nodes. Google reportedly sets α approx 0.85.

This modification ensures that every page in the network is reachable from every other page, making the chain irreducible. The chain is also aperiodic due to random teleportation breaking cycles. Furthermore, the chain is positive recurrent, and thus, by the Fundamental Theorem of Markov Chains, a unique stationary distribution exists.

The Google Matrix for the example in Figure-1 allows us to calculate a stationary distribution and thus rank the importance of the pages (using α approx 0.85):

$$G = \alpha S + (1-\alpha) J = \alpha S + \frac{(1-\alpha)}{n} \mathbf{1} \mathbf{1}^T + \alpha (1-\alpha) \mathbf{1} \mathbf{1}^T$$

References:

- [1] David C. Long, *Mathematical Foundations of PageRank*, <https://www.math.ucdavis.edu/~long/pagerank.pdf>.
- [2] National Science and Media Museum, *A Short History of the Internet*, December 2020, <https://www.scienceandmediamuseum.org.uk/object/short-history-internet>.
- [3] Webspam.org, *SEO Spam: What is Spamdexing?*, <https://webspam.org/seo-spam-what-is-spamdexing/>.
- [4] L. Page, S. Brin, R. Motwani, and T. Winograd, *The PageRank Citation Ranking to the Web*, Stanford University, Technical Report 1999-66, 1999, <http://ilpubs.stanford.edu/publications/techreports/1999-66.pdf>.
- [5] S. Carreres, *Mathematics for Industry Case Study 1 - Google PageRank*, 2025.
- [6] A. Brandt, *Directed Graphs*, <https://www.csd.uwo.ca/~brandt/DiscreteStructures/Chapter7/directedgraphs.html>.
- [7] Georgia Institute of Technology, *Stochastic Matrices*, <https://textbooks.math.gatech.edu/stochastic-matrices.html>.

Figure 7: Overleaf Word Count